

Quality-Critical Azure Lakehouse

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From batch and streaming telemetry to governed, traceable data products.



Trust has to survive the whole data path

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Operations needs to know whether a batch stayed within limits, what changed, which records failed, and where every KPI originated.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

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02 / 32

Two ingestion modes. One governed lakehouse.

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ADF lands deterministic batch files while Event Hubs feeds bounded telemetry into checkpointed Structured Streaming.



The evidence spans the engineering lifecycle

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Provision, secure, ingest, transform, govern, orchestrate, break, repair, measure, monitor, publish, and tear down.

IMPLEMENTATION EVIDENCE

Bicep infrastructure as code**ADLS Gen2 medallion storage****Azure Data Factory batch ingestion****Event Hubs bounded streaming****PySpark and Delta Lake transformations****Lakeflow pipelines and Jobs****Unity Catalog governance and lineage****CDC and SCD Type 2 history****Failure repair and idempotency proof**

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04 / 32

Trial-only infrastructure with fail-closed cost gates

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Subscription-scope Bicep creates only the isolated resource group and expected Azure service set.

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05 / 32

Managed identity is the default trust path

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ADF, the Access Connector, and deployment automation receive scoped access without long-lived cloud passwords.

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06 / 32

ADLS Gen2 separates landing, checkpoints, quarantine, and evidence

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Hierarchical namespace, TLS 1.2, disabled shared keys, and deliberate paths make state and ownership clear.

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07 / 32

ADF ingests six commit-pinned source files

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A parameterized ForEach and Copy pattern preserves file-level execution and transfer evidence.

IMPLEMENTATION EVIDENCE



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08 / 32

Runs are reconciled by files, rows, bytes, and status

PRODUCTION_BLUEPRINT

The final evidence binds the ADF pipeline and activity output to landed ADLS objects.

EXECUTION EVIDENCE PENDING



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EXECUTION RECEIPT

The platform artifact will promote this claim only after execution.

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09 / 32

Twenty thousand deterministic telemetry messages

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The producer is bounded, seed-controlled, receipt-producing, and excludes its connection material from output.

IMPLEMENTATION EVIDENCE



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10 / 32

Explicit schema, checkpoint, metadata, append semantics

PRODUCTION_BLUEPRINT

Streaming proof combines Event Hubs ingress, progress metrics, checkpoint existence, and Bronze Delta counts.

EXECUTION EVIDENCE PENDING**IMPLEMENTATION**

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11 / 32

Governance belongs inside the data path

PRODUCTION_BLUEPRINT

Catalog-qualified tables, storage credentials, external locations, ownership, grants, tags, and policies create inspectable control.

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12 / 32

Raw does not mean unaccountable

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Bronze preserves source file and system, ingest and event time, pipeline run, record hash, and schema version.

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13 / 32

Warn, quarantine, drop, and fail are different decisions

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Lakeflow expectations publish warning metrics, route bad records, and enforce one reserved hard contract failure.

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14 / 32

Bad records stay explainable

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Deterministic reason codes retain invalid keys, timestamps, ranges, units, duplicates, and unknown references for inspection.

IMPLEMENTATION EVIDENCE**IMPLEMENTATION**

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15 / 32

Conformance creates a dependable analytical boundary

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Explicit types, keys, units, domain rules, joins, deduplication, and idempotency shape validated tables.

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16 / 32

History is preserved, not overwritten

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AUTO CDC is preferred, with a deterministic Delta MERGE fallback and one-current-version proof.

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17 / 32

Six outputs with explicit business grain

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Two facts, three dimensions, and a site-product-date KPI table each document keys, lineage, and validation SQL.

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fact_batch_quality

fact_cold_chain_excursion

dim_batch_history

dim_site

dim_product

kpi_quality_summary

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18 / 32

The DAG exposes dependencies and recovery boundaries

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Parallel Bronze paths converge at quality, Silver and SCD2 run in parallel, then Gold, validation, and receipt publication complete.

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19 / 32

Gold should be traceable to Bronze

PRODUCTION_BLUEPRINT

Table-name-based Unity Catalog operations are designed to generate actual source-to-product lineage.

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20 / 32

Traceability reaches the field level

PRODUCTION_BLUEPRINT

The selected evidence path follows a quality or temperature field from source through conformance into a Gold measure.

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21 / 32

A reserved hard violation breaks the real task

PRODUCTION_BLUEPRINT

The incident is isolated, deterministic, and captured with the platform diagnostic rather than simulated in presentation code.

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22 / 32

The failure points to one violated contract

PRODUCTION_BLUEPRINT

The root-cause note connects the malformed schema version to the affected quality task and excludes unrelated upstream work.

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23 / 32

Repair the failed path, preserve completed work

PRODUCTION_BLUEPRINT

Lakeflow repair is used where the Trial runtime supports it; otherwise the exact limitation remains explicit.

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24 / 32

Recovery is verified by content, not appearance

PRODUCTION_BLUEPRINT

Counts, duplicates, aggregates, and deterministic clean-versus-recovered hashes must all reconcile.

EXECUTION EVIDENCE PENDING



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25 / 32

A separate five-million-row fixture creates an honest test

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Three baseline runs record wall time, stages, tasks, shuffle, spill, partitions, max duration, and p75 duration.

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26 / 32

Broadcast the small dimension; remove needless repartitioning

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Three optimized runs on the same compute are compared by medians, with the measured result preserved even if slower.

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27 / 32

Operational evidence crosses service boundaries

PRODUCTION_BLUEPRINT

Diagnostics ingest before execution; KQL uses the tables that actually appear and an alert is only called fired if it fires.

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28 / 32

Cost labels state exactly what billing can prove

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Snapshots use actual, current, estimate, or pending-settlement labels while \$15 and \$20 gates control retries and teardown.

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29 / 32

Every release travels through validation

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Python, Spark, Bicep, bundle, content, accessibility, secret, metadata, PDF, and Pages checks run before publication.

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30 / 32

The lifecycle ends with authoritative absence

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Only the isolated resource groups and Part 4 budget are deleted, then Azure readback is polled until absence is confirmed.

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31 / 32

Inspect the complete engineering journey

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Every major claim connects to platform capture, implementation, receipt, validation, time, commit, and hash across the repository, live case study, evidence explorer, and teardown record.

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32 / 32

<https://smailliwhtes.github.io/quality-critical-azure-lakehouse/>